

Appendix: Accuracy, Precision and Uncertainty - ERROR ANALYSIS

How tall are you? How old are you? When you answered these everyday questions, you probably did it in round numbers such as "five foot, six inches" or "nineteen years, three months." But how true are these answers? Are you exactly 5' 6" tall? Probably not. You estimated your height at 5' 6" and just reported two significant figures. Typically, you round your height to the nearest inch, so that your actual height falls somewhere between 5' 5½" and 5' 6½" tall, or 5' 6" $\pm$ ½". This $\pm$ ½" is the **uncertainty**, and it informs the reader of the precision of the **value** 5' 6".

What is uncertainty?

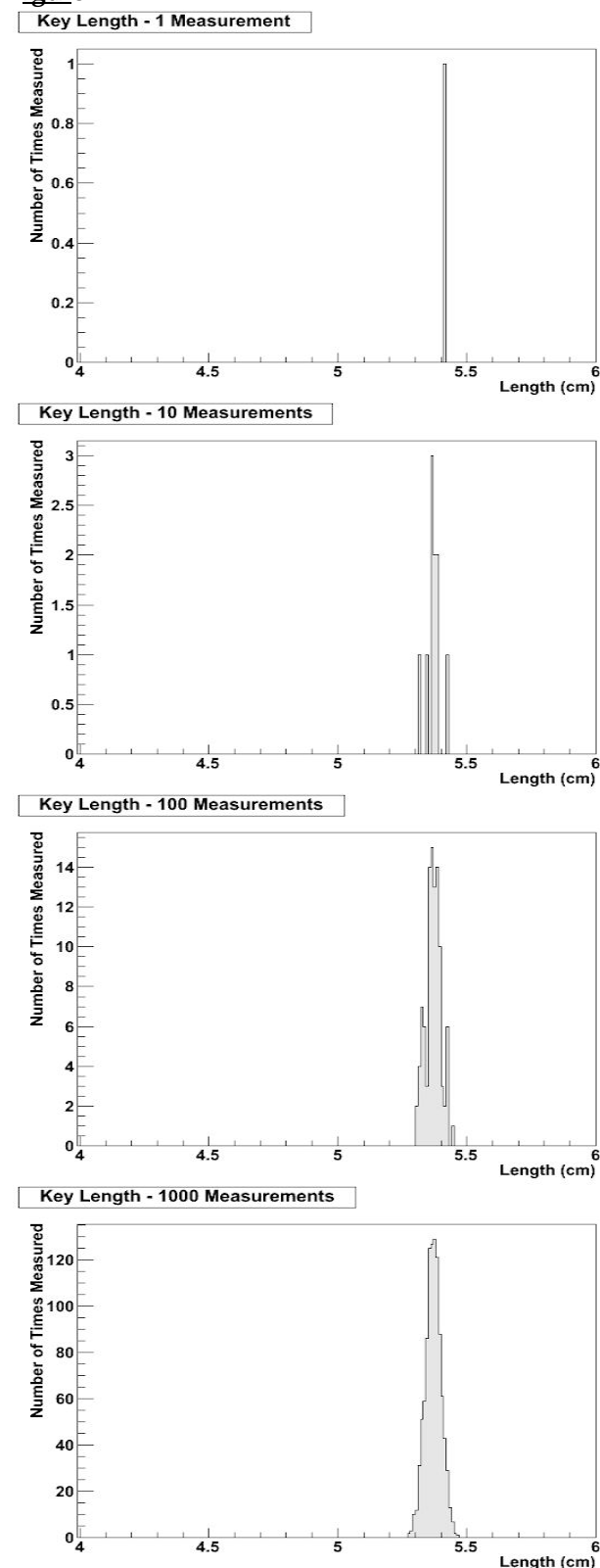
Whenever you measure something, there is always some uncertainty. There are two categories of uncertainty: **systematic** and **random**.

(1) **Systematic uncertainties** are those that consistently cause the value to be too large or too small. Systematic uncertainties include such things as reaction time, inaccurate meter sticks, optical parallax and miscalibrated balances. In principle, systematic uncertainties can be eliminated if you know they exist.

(2) **Random uncertainties** are variations in the measurements that occur without a predictable pattern. If you make precise measurements, these uncertainties arise from the estimated part of the measurement. Random uncertainty can be reduced, but never eliminated. We need a technique to report the contribution of this uncertainty to the measured value.

Uncertainties cause every measurement you make to be **distributed**. For example, the key in Figure 2 is approximately 5.37cm long. For the sake of argument, pretend that it is exactly 5.37cm long. If you measure its length many times, you expect that most of the measurements will be close to, but not exactly, 5.37cm, and that there will be a few measurements much more than or much less than 5.37cm. This effect is due to random uncertainty. You can never know how accurate any single measurement is, but you expect that many measurements will cluster around the real length, so you can take the average as the "real" length, and more measurements will give you a better answer; see Figure 1.

Figure 1



You must be very careful to estimate or eliminate (by other means) systematic uncertainties well because they cannot be eliminated in this way; they would just shift the distributions in Figure 1 left or right.

Roughly speaking, the average or “center” of the distribution is the “measurement,” and the width or “deviation” of the distribution is the random uncertainty.

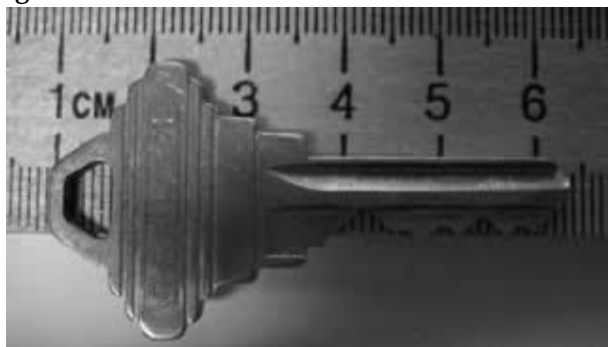
How do I determine the uncertainty?

This Appendix will discuss three basic techniques for determining the uncertainty: **estimating the uncertainty**, measuring the **average deviation**, and finding the **uncertainty in a linear fit**. Which one you choose will depend on your situation, your available means of measurement, and your need for precision. If you need a precise determination of some value, and you are measuring it directly (e.g., with a ruler or thermometer), the best technique is to measure that value several times and use the average deviation as the uncertainty. Examples of finding the average deviation are given below.

How do I estimate uncertainties?

If time or experimental constraints make repeated measurements impossible, then you will need to estimate the uncertainty. When you estimate uncertainties you are trying to account for anything that might cause the measured value to be different if you were to take the measurement again. For example, suppose you were trying to measure the length of a key, as in Figure 2.

Figure 2



If the true value were not as important as the magnitude of the value, you could say that the key's length was 5 cm, give or take 1 cm. This is a crude estimate, but it may be acceptable. A better estimate of the key's length, as you saw in Appendix A, would

be 5.37 cm. This tells us that the worst our measurement could be off is a fraction of a mm. To be more precise, we can estimate it to be about a third of a mm, so we can say that the length of the key is 5.37 ± 0.03 cm.

Another time you may need to estimate uncertainty is when you analyze video data. Figures 3 and 4 show a ball rolling off the edge of a table. These are two consecutive frames, separated in time by $1/30$ of a second.

Figure 3

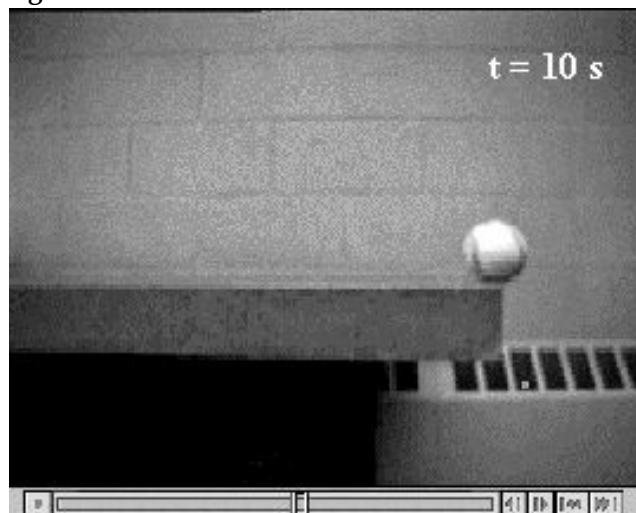
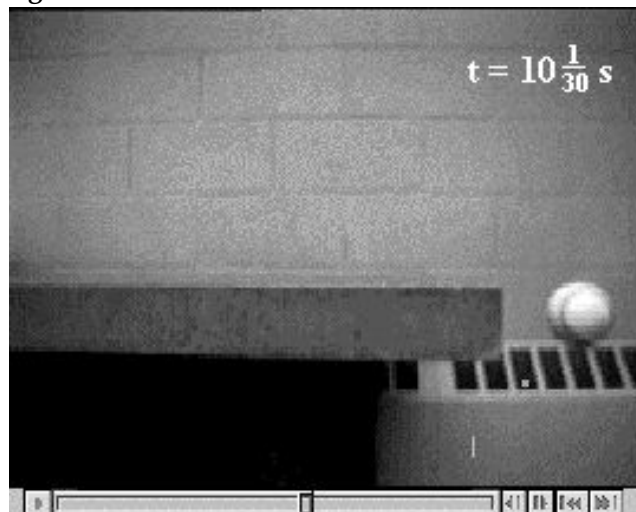


Figure 4



The exact moment the ball left the table lies somewhere between these frames. We can estimate that this moment occurs midway between them ($t = 10 \frac{1}{60} s$). Since it must occur at some point between them, the worst our estimate could be off by

is $\frac{1}{60} s$. We can therefore say the time the ball leaves the table is $t = 10 \frac{1}{60} \pm \frac{1}{60} s$.

How do I find the average deviation?

If estimating the uncertainty is not good enough for your situation, you can experimentally determine the uncertainty by making several measurements and calculating the average deviation of those measurements. To find the average deviation: (1) Find the average of all your measurements; (2) Find the absolute value of the difference of each measurement from the average (its deviation); (3) Find the average of all the deviations by adding them up and dividing by the number of measurements. Of course you need to take enough measurements to get a distribution for which the average has some meaning.

In example 1, a class of six students was asked to find the mass of the same penny using the same balance. In example 2, another class measured a different penny using six different balances. Their results are listed below:

Class 1: Penny A massed by six different students on the same balance.

Mass (grams)
3.110
3.125
3.120
3.126
3.122
<u>3.120</u>
3.121 average.
The deviations are: 0.011g, 0.004g, 0.001g, 0.005g, 0.001g, 0.001g
Sum of deviations: 0.023g
Average deviation: (0.023g)/6 = 0.004g
Mass of penny A: $3.121 \pm 0.004g$

Class 2: Penny B massed by six different students on six different balances

Mass (grams)
3.140
3.133
3.144
3.118
3.126
<u>3.125</u>
3.131 average

The deviations are: 0.009g, 0.002g, 0.013g, 0.013g, 0.005g, 0.006g
Sum of deviations: 0.048g
Average deviation: (0.048g)/6 = 0.008g
Mass of penny B: $3.131 \pm 0.008g$

Finding the Uncertainty in a Linear Fit

Sometimes, you will need to find the uncertainty in a linear fit to a large number of measurements. The most common situation like this that you will encounter is fitting position or velocity with respect to time from MotionLab.

When you fit a line to a graph, you will be looking for the "best fit" line that "goes through the middle" of the data; see the appendix about graphs for more about this procedure. To find the uncertainty, draw the lines with the greatest and least slopes that still roughly go through the data. These will be the upper and lower limits of the uncertainty in the slope. These lines should also have lesser and greater y-intercepts than the "best fit" line, and they define the lower and upper limits of the uncertainty in the y-intercept.

Note that when you do this, the uncertainties above and below your "best fit" values will, in general, **not** be the same; this is different than the other two methods we have presented.

For example, in Figure 5, the y-intercept is $4.25 \pm 2.75/-2.00$, and the slope is $0.90 \pm 0.20/-0.25$.

Figure 5a

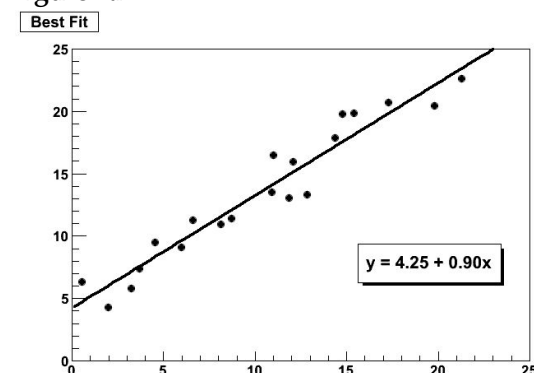


Figure 5b

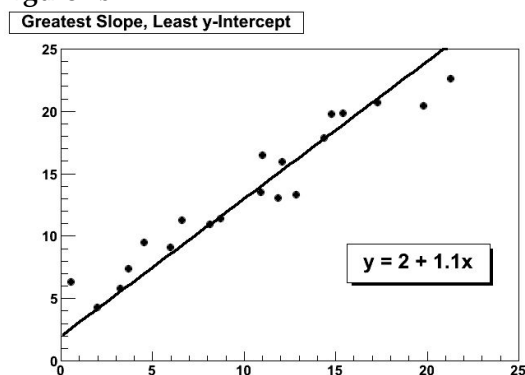
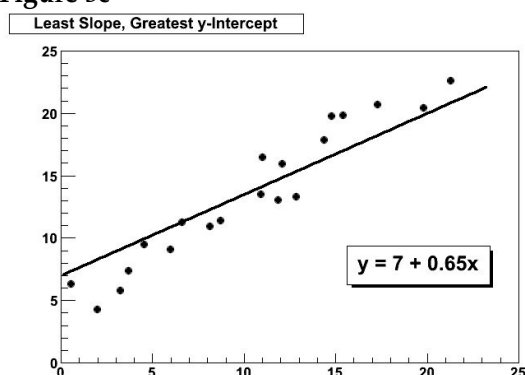


Figure 5c

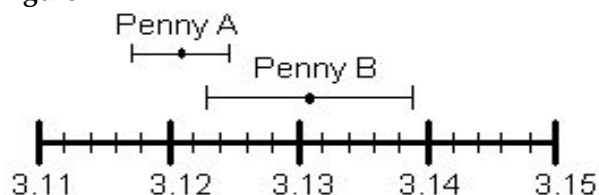


However you choose to determine the uncertainty, you should always state your method clearly in your report.

How do I know if two values are the same?

Go back to the pennies. If we compare only the average masses of the two pennies we see that they are different. But now include the uncertainty in the masses. For penny A, the most likely mass is somewhere between 3.117g and 3.125g. For penny B, the most likely mass is somewhere between 3.123g and 3.139g. If you compare the ranges of the masses for the two pennies, as shown in Figure 6, they just overlap. Given the uncertainty in the masses, we are able to conclude that the masses of the two pennies could be the same. If the range of the masses did not overlap, then we ought to conclude that the masses are probably different.

Figure 6



Mass of pennies (in grams) with uncertainties

An important application of this is determining agreement between experimental and theoretical values. If you use a formula to generate a theoretical value of some quantity and use the method below to generate the uncertainty in the calculation, and if you generate an experimental value of the same quantity by measuring it and use the method above to generate the uncertainty in the measurement, you can compare the two values in this way. If the ranges overlap, then the theoretical and experimental values agree. If the ranges do not overlap, then the theoretical and experimental values do not agree.

Which result is more precise?

Suppose you use a meter stick to measure the length of a table and the width of a hair, each with an uncertainty of 1 mm. Clearly you know more about the length of the table than the width of the hair. Your measurement of the table is very precise but your measurement of the width of the hair is rather crude. To express this sense of precision, you need to calculate the percentage uncertainty. To do this, divide the uncertainty in the measurement by the value of the measurement itself, and then multiply by 100%. For example, we can calculate the precision in the measurements made by class 1 and class 2 as follows:

Precision of Class 1's value:

$$(0.004 \text{ g} \div 3.121 \text{ g}) \times 100\% = 0.1 \%$$

Precision of Class 2's value:

$$(0.008 \text{ g} \div 3.131 \text{ g}) \times 100\% = 0.3 \%$$

Class 1's results are more precise. This should not be surprising since class 2 introduced more uncertainty in their results by using six different balances instead of only one.

Which result is more accurate?

Accuracy is a measure of how your measured value compares with the real value. Imagine that class 2 made the measurement again using only one balance. Unfortunately, they chose a balance that was poorly

calibrated. They analyzed their results and found the mass of penny B to be 3.556 ± 0.004 g. This number is more precise than their previous result since the uncertainty is smaller, but the new measured value of mass is very different from their previous value. We might conclude that this new value for the mass of penny B is different, since the range of the new value does not overlap the range of the previous value. However, that conclusion would be **wrong** since our uncertainty has not taken into account the inaccuracy of the balance. To determine the accuracy of the measurement, we should check by measuring something that is known. This procedure is called calibration, and it is absolutely necessary for making accurate measurements.

Be cautious! It is possible to make measurements that are extremely precise and, at the same time, grossly inaccurate.

How can I do calculations with values that have uncertainty?

When you do calculations with values that have uncertainties, you will need to estimate (by calculation) the uncertainty in the result. There are mathematical techniques for doing this, which depend on the statistical properties of your measurements. A very simple way to estimate uncertainties is to find the *largest possible uncertainty* the calculation could yield. **This will always overestimate the uncertainty of your calculation**, but an overestimate is better than no estimate or an underestimate. The method for performing arithmetic operations on quantities with uncertainties is illustrated in the following examples:

Addition:

$$(3.131 \pm 0.008 \text{ g}) + (3.121 \pm 0.004 \text{ g}) = ?$$

First, find the sum of the values:

$$3.131 \text{ g} + 3.121 \text{ g} = 6.252 \text{ g}$$

Next, find the largest possible value:

$$3.139 \text{ g} + 3.125 \text{ g} = 6.264 \text{ g}$$

The uncertainty is the difference between the two:

$$6.264 \text{ g} - 6.252 \text{ g} = 0.012 \text{ g}$$

Answer: 6.252 ± 0.012 g.

Note: This uncertainty can be found by simply adding the individual uncertainties:

$$0.004 \text{ g} + 0.008 \text{ g} = 0.012 \text{ g}$$

Multiplication:

$$(3.131 \pm 0.013 \text{ g}) \times (6.1 \pm 0.2 \text{ cm}) = ?$$

First, find the product of the values:

$$3.131 \text{ g} \times 6.1 \text{ cm} = 19.1 \text{ g-cm}$$

Next, find the largest possible value:

$$3.144 \text{ g} \times 6.3 \text{ cm} = 19.8 \text{ g-cm}$$

The uncertainty is the difference between the two:

$$19.8 \text{ g-cm} - 19.1 \text{ g-cm} = 0.7 \text{ g-cm}$$

Answer: 19.1 ± 0.7 g-cm.

Note: The percentage uncertainty in the answer is the sum of the individual percentage uncertainties:

$$\frac{0.013}{3.131} \times 100\% + \frac{0.2}{6.1} \times 100\% = \frac{0.7}{19.1} \times 100\%$$

Subtraction: $(3.131 \pm 0.008 \text{ g}) - (3.121 \pm 0.004 \text{ g}) = ?$ First, find the difference of the values: $3.131 \text{ g} - 3.121 \text{ g} = 0.010 \text{ g}$ Next, find the largest possible difference: $3.139 \text{ g} - 3.117 \text{ g} = 0.022 \text{ g}$ The uncertainty is the difference between the two: $0.022 \text{ g} - 0.010 \text{ g} = 0.012 \text{ g}$ Answer: $0.010 \pm 0.012 \text{ g}$. <i>Note: This <u>uncertainty</u> can be found by simply adding the <u>individual uncertainties</u>:</i> $0.004 \text{ g} + 0.008 \text{ g} = 0.012 \text{ g}$ <i>Notice also, that zero is included in this range, so it is possible that there is no difference in the masses of the pennies, as we saw before.</i>	Division: $(3.131 \pm 0.008 \text{ g}) \div (3.121 \pm 0.004 \text{ g}) = ?$ First, divide the values: $3.131 \text{ g} \div 3.121 \text{ g} = 1.0032$ Next, find the largest possible value: $3.139 \text{ g} \div 3.117 \text{ g} = 1.0071$ The uncertainty is the difference between the two: $1.0071 - 1.0032 = 0.0039$ Answer: 1.003 ± 0.004 <i>Note: The <u>percentage uncertainty</u> in the answer is the sum of the <u>individual percentage uncertainties</u>:</i> $\frac{0.008}{3.131} \times 100\% + \frac{0.004}{3.121} \times 100\% = \frac{0.0039}{1.0032} \times 100\%$ <i>Notice also, the largest possible value for the numerator and the smallest possible value for the denominator gives the largest result.</i>
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The same ideas can be carried out with more complicated calculations. Remember this will always give you an overestimate of your uncertainty. There are other calculation techniques, which give better estimates for uncertainties. If you wish to use them, please discuss it with your instructor to see if they are appropriate.

These techniques help you estimate the random uncertainty that always occurs in measurements.

They will not help account for mistakes or poor measurement procedures. There is no substitute for taking data with the utmost of care. A little forethought about the possible sources of uncertainty can go a long way in ensuring precise and accurate data.

PRACTICE EXERCISES:

B-1. Consider the following results for different experiments. Determine if they agree with the accepted result listed to the right. Also calculate the precision for each result.

a) $g = 10.4 \pm 1.1 \text{ m/s}^2$

$g = 9.8 \text{ m/s}^2$

b) $T = 1.5 \pm 0.1 \text{ sec}$

$T = 1.1 \text{ sec}$

c) $k = 1368 \pm 45 \text{ N/m}$

$k = 1300 \pm 50 \text{ N/m}$

B-2. The area of a rectangular metal plate was found by measuring its length and its width. The length was found to be $5.37 \pm 0.05 \text{ cm}$. The width was found to be $3.42 \pm 0.02 \text{ cm}$. What is the area and the average deviation?

B-3. Each member of your lab group weighs the cart and two mass sets twice. The following table shows this data. Calculate the total mass of the cart with each set of masses and for the two sets of masses combined.

Cart (grams)	Mass set 1 (grams)	Mass set 2 (grams)
201.3	98.7	95.6
201.5	98.8	95.3
202.3	96.9	96.4
202.1	97.1	96.2
199.8	98.4	95.8
200.0	98.6	95.6